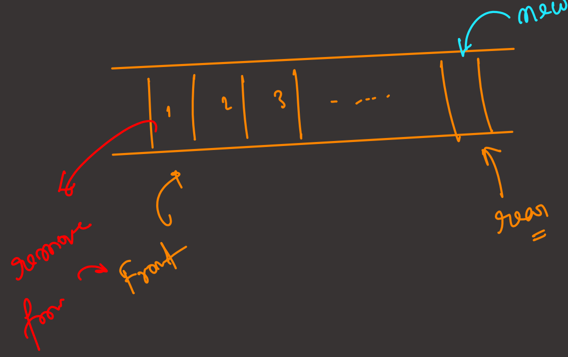
# Queue    First in first Out

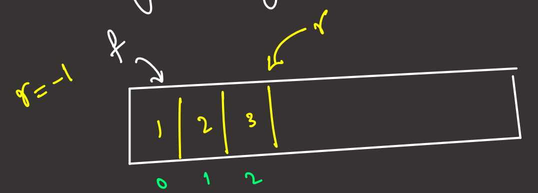
insertion order = deletion order




- insert / add / enqueue $O(1)$
- deletion / remove / dequeue . $O(1)$ Deque ← DS
- Peak () → Front $O(1)$

## Implementation of Queue

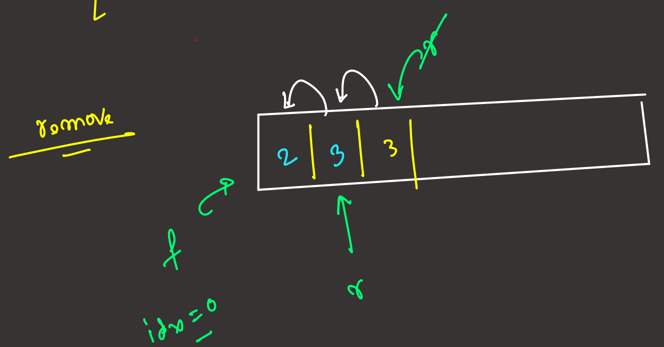
(i) using Array



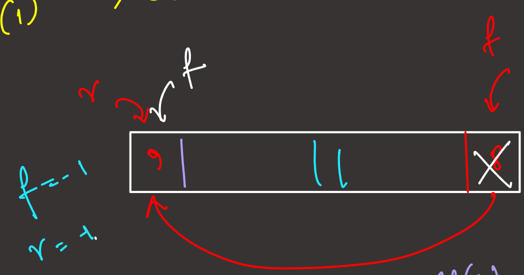

$O(1)$

insert

[1, 2, 3]




$O(n)$ → $O(1)$ → Circular Queue

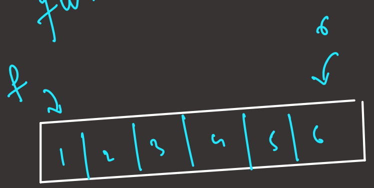



add(9)

add (1)
add (2)
add (3)
remove () → 1

rear = (rear + 1) % size
front = (front + 1) % size

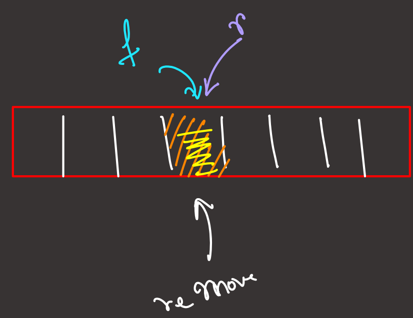
full → { r - f = size-1 }

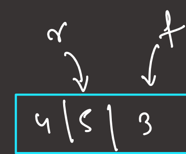



(r+1) % size = f

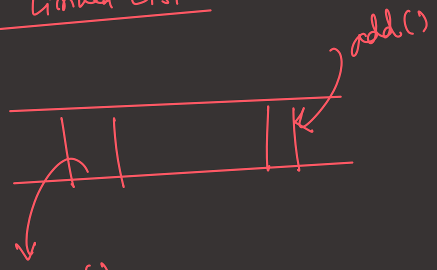
condition of empty

{ f = -1
  r = -1 }



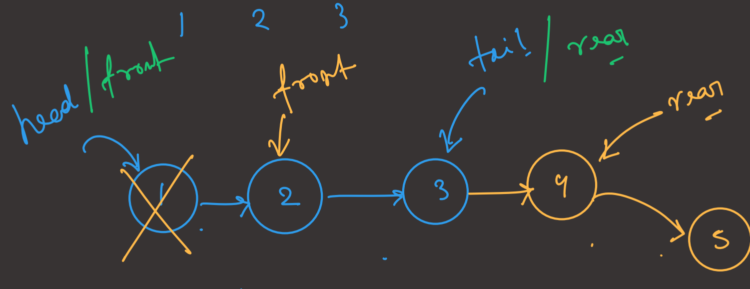
## Queue using Linked List




$O(1)$ add()

$O(1)$ remove()

$O(1)$ peek()




## Queue Using JCF

Queue ← interface.

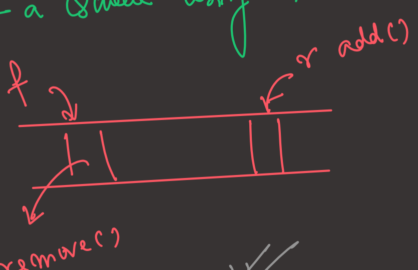
✓✓ Queue < Integer > $q_1$ = new LinkedList <>() ;

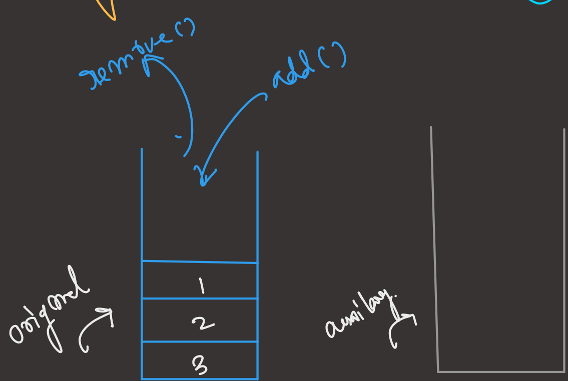
Queue < Integer > $q_2$ = new ArrayDeque <>() ;

[1 2 3]

## Question

implement a Queue using two stack




H.W

way 1

add → $O(1)$

and remove → $O(n)$

✓✓ or way 2

add → $O(n)$

remove → $O(1)$

HW

✓✓ way 1

push → $O(1)$

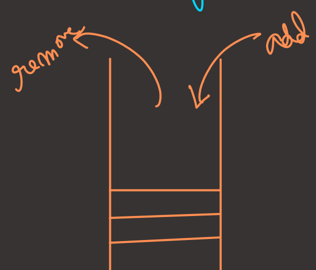
pop → $O(n)$

way 2

push → $O(n)$

pop → $O(1)$

## Stack Using two Queue

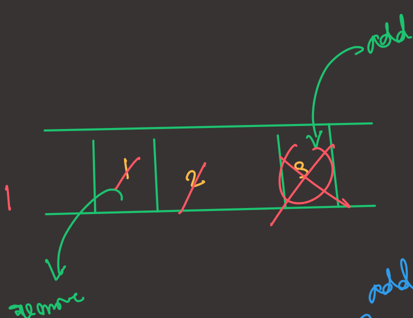



push

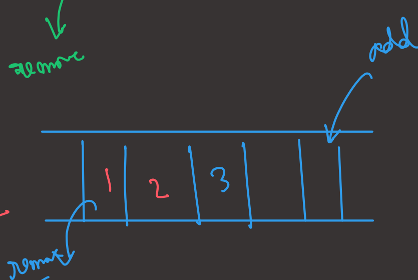
[1 2 3]

pop [3 2 1]

$q_1$




$q_2$



$T.C = O(n)$

↑

Pop, Peek

```
Pop() { top = -1
    if (!q1.isEmpty()) {
        while (!q1.isEmpty()) {
            top = q1.remove();
            if (q1.isEmpty()) {
                break;
            }
            q2.add(top);
        }
    }
    else {
        while (!q2.isEmpty()) {
            top = q2.remove();
            if (q2.isEmpty()) {
                break;
            }
            q1.add(top);
        }
    }
    return top
}
```

**Question**

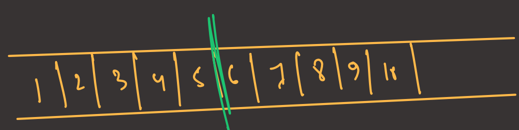
interleave 2 halves of a Queue. (even length)

given: 1 2 3 4 5 | 6 7 8 9 10

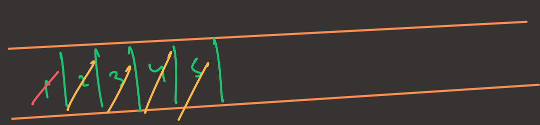
o/p: 1 6 2 7 3 8 4 9 5 10

upto q.size()/2 from initial queue remove and put it into the new queue

initial q




new q




```
while (!newq.isEmpty()) {
    initialq.add(newq.remove());
    initialq.add(initialq.remove());
}
```

initial